

Lecture 5 Notes - Simple Linear Regression

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- **Reading:** Chapter 2 – Shumway and Stoffer

1 Review

1.1 Notes on covariance

To add to your materials from last week, I've created some visualizations.

2 Today

2.1 Simple linear regression

Let's say we want to learn the relationship between two variables y and x - the goal is to predict y given x . For example, we might want to predict the height of an adult man (y) given the height of his father (x). y is called the response variable or dependent variable, x is the covariate or independent variable. We will start with the more general scenario and then extend this concept specifically to time series.

In simple linear regression, we are predicting y given one covariate x . In the case of multiple x , say $\{x_1, \dots, x_p\}$, this is called multiple regression.

For the simple case, we are predicting y from one covariate x . For example:

$$y = \beta_0 + \beta_1 x + \epsilon$$

β_0 and β_1 are parameters that we will estimate from the data, where β_0 is the intercept, which corresponds to the value of y when $x = 0$, and β_1 represents the change in y when x changes by one unit.

We could observe, for example, pairs of data $(x_1, y_1), \dots, (x_n, y_n)$ where each of these samples is a pair of fathers and sons. We then might write:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i.$$

We can then estimate our $\hat{\beta}_0$ and $\hat{\beta}_1$ from the data, then use these to predict the value of the response variable $\hat{y}$ (height of a new adult man) for a new covariate x . We will talk about how to do this using the Python library `statsmodels`, but we will also talk about how to derive the solution mathematically.

For time series regression, we often have some observed data $y_1, \dots, y_n$ that are observations for a single variable y . In this case, we may try two main ways of predicting y :

1. **Use time as a covariate** (many of the examples we've seen already do this – for example, the DJIA data, fMRI data, population data over time)

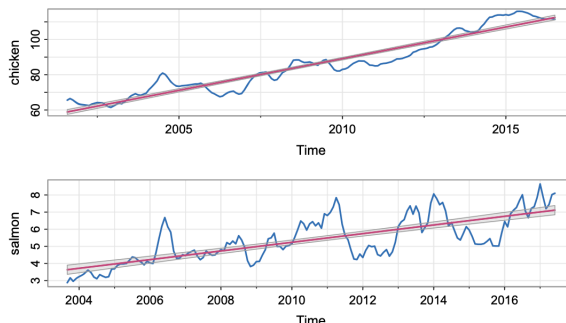


Fig. 2.1. The price of chicken in US cents per pound, 2001–2016 (top). The export price of farm-bred Norwegian salmon in US dollars per kilogram, 2003–2017 (bottom). Each plot shows the data along with the fitted linear trend and 95% pointwise confidence intervals.

[resume] **Use lagged version of y as the covariate.** For this case, we might take $x_i = y_{i-1}$. In this case, we are predicting a current value based on a prior time point. This is called Lagged Regression or AutoRegression.

2.2 How do we estimate β_0 and β_1 ?

There are a couple of ways to estimate β_0 and β_1 . Standard libraries such as `statsmodels` use the least squares method. In this method, we minimize the error sum of squares:

$$Q = \sum_{t=1}^n (y - \hat{y})^2 = \sum_{t=1}^n (y - \beta_0 - \beta_1 x)^2 \quad (1)$$

2.2.1 Ordinary Least Squares (OLS)

In ordinary least squares, we can solve for:

$\min_{\beta_0, \beta_1} \mathbb{E}[(y - \beta_0 - \beta_1 x)^2]$, which will be the best fitting line at the population level.

Solving this gives:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \text{ and } \hat{\beta}_1 = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

where

$$\bar{y} = \frac{y_1 + \dots + y_n}{n} \text{ and } \bar{x} = \frac{x_1 + \dots + x_n}{n}$$

Let's look at what this looks like in `statsmodels` with an example.